Assignment 1: Bike Rental Demand Prediction Using Linear Regression

Uday Parmar(261081860) Shahmeer Sajid(261073334) Danyal Valika(261036640)

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1 Abstract

In Assignment 1, we used the `day.csv` dataset to build a linear regression model predicting the total number of bikes rented per day. The dataset was cleaned and preprocessed, followed by exploratory data analysis to identify relationships among covariates. Feature scaling and one-hot encoding were applied according to variable types.

Model performance was progressively improved through EDA-driven feature engineering. Non-linear patterns were identified, particularly involving temperature and its interactions with humidity, windspeed, season, and calendar variables. Polynomial and interaction terms were introduced in successive stages to better capture these relationships. Improvements were evaluated by comparing training and test errors across feature sets.

Each modification led to measurable performance gains, confirmed through percentage reductions in error. The final model achieved a training RMSE of approximately ± 583 bikes and a test RMSE of approximately ± 672 bikes, demonstrating that carefully motivated feature engineering can significantly enhance linear regression performance on real-world data.

2 Introduction

The dataset used in this project, `day.csv`, contains daily bike rental counts along with covariates describing weather conditions, seasonality, year, and holiday status. It consists of 731 observations spanning two years and includes both continuous and categorical variables.

The dataset contained no missing values, eliminating the need for imputation. Given the continuous nature of the target variable, linear regression was chosen as the baseline model. The primary objective was to improve predictive accuracy while preserving interpretability and avoiding data leakage or overfitting.

A central focus of the project was understanding relationships between variables through exploratory analysis. By systematically adding non-linear and interaction features guided by observed patterns, we aimed to improve model performance while remaining within the linear regression framework.

3 Data Preprocessing and Exploration

Initial inspection of the dataset confirmed the absence of missing values, malformed entries, or obvious anomalies. Features that could cause data leakage or were not available at prediction time were removed prior to modeling. Categorical variables, including season, weather situation, month, weekday, year, holiday, and working day, were transformed using one-hot encoding to avoid imposing artificial ordinal structure. Continuous variables (temperature, feels-like temperature, humidity, and windspeed) were standardized to zero mean and unit variance to improve numerical stability and ensure comparable coefficient scales when solving the normal equations.

Exploratory analysis of the target variable showed that daily bike rental counts are approximately symmetrically distributed, with the mean close to the median and no extreme outliers (Figure 1). This distribution suggested that no trimming or transformation of the target variable was necessary.

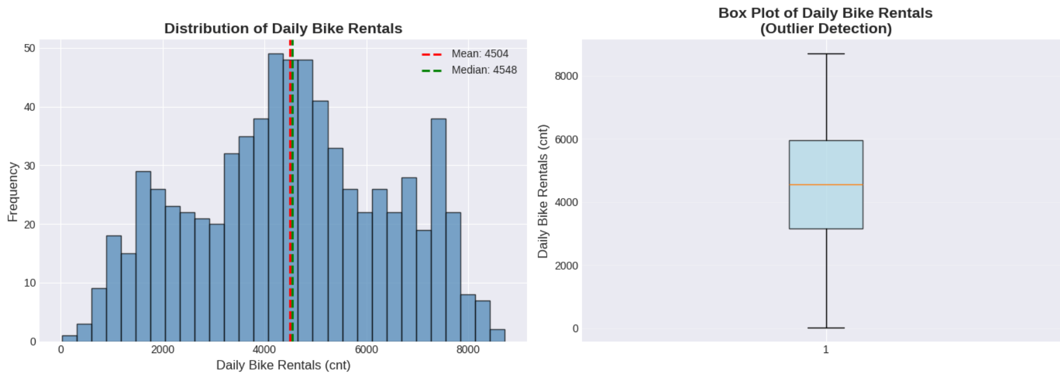


Figure 1: Distribution and box plot of daily bike rentals, showing a roughly symmetric distribution with no extreme outliers.

Distributions of continuous features are shown in Figure 2. Temperature and feels-like temperature exhibit approximately bell-shaped distributions, while humidity shows moderate skewness and windspeed is right-skewed. These differing scales and spreads motivated the use of standardization and later informed the introduction of polynomial feature expansions.

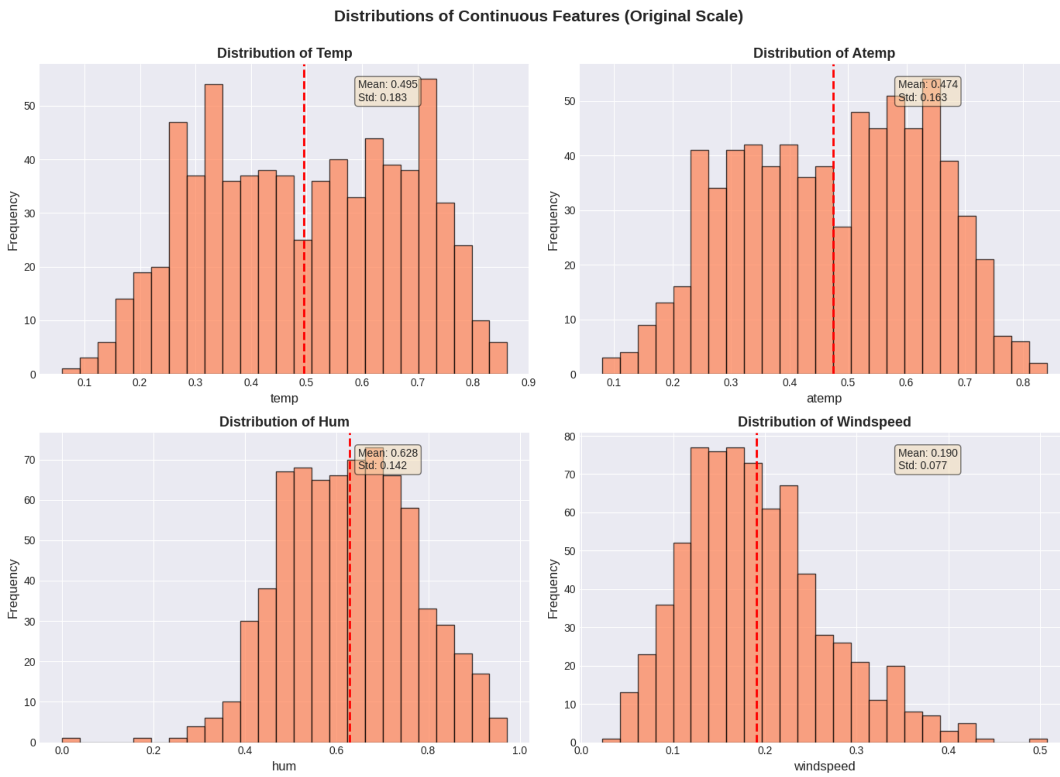
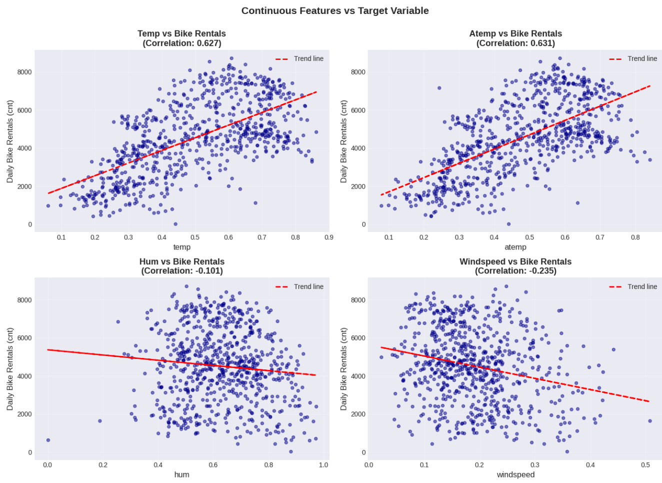
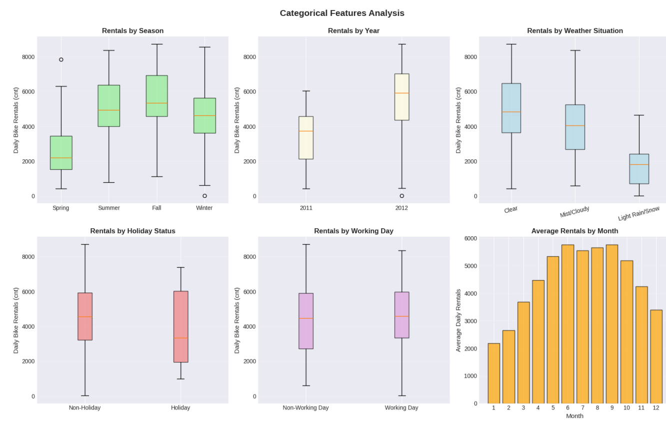


Figure 2: Distributions of continuous features on their original scales.

To examine relationships between predictors and the target variable, we visualized both continuous and categorical features (Figure 3). Continuous features related to temperature show strong positive associations with bike rentals, while humidity and windspeed display weaker, predominantly negative relationships. Notably, the temperature-related plots exhibit curved patterns rather than purely linear trends, motivating the use of polynomial terms. Categorical features reveal substantial variation across seasons, years, and weather conditions: Fall and summer have the highest median rentals, usage in 2012 exceeds that of 2011, and clear weather days consistently outperform rainy or snowy days.



(a) Continuous features plotted against daily bike rentals.



(b) Bike rentals grouped by categorical variables.

Figure 3: Exploratory analysis of continuous and categorical features.

Finally, a correlation heatmap (Figure 4) highlights relationships among features and the target variable. Temperature and feels-like temperature exhibit strong collinearity, while season and year show moderate positive correlations with bike rentals. Calendar variables such as weekday and holiday are more weakly associated. These observations informed subsequent feature engineering choices and are discussed further in Section 6.

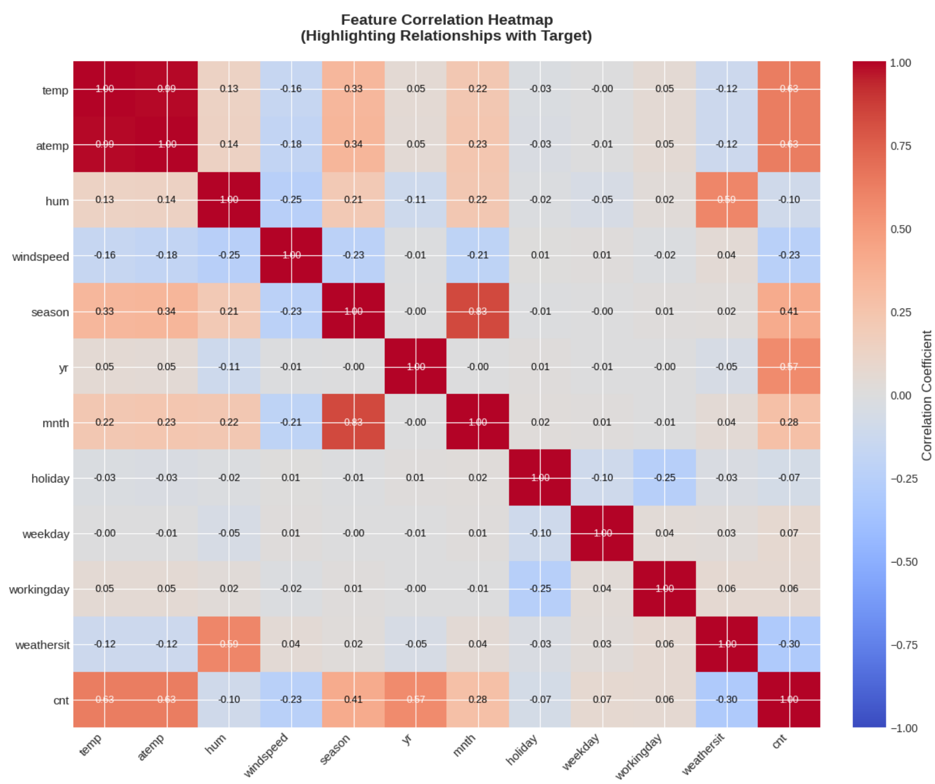


Figure 4: Correlation heatmap highlighting relationships between features and the target variable.

4 Methods

4.1 Linear Regression Implementation

We implemented linear regression from scratch in Python using NumPy and the closed-form solution (normal equation). Given a feature matrix $\mathbf{X}$ and target vector $\mathbf{y}$, the optimal weight vector is computed as

$$\mathbf{w} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}.$$

For numerical stability, we solved the equivalent linear system using `np.linalg.solve` rather than explicitly computing the matrix inverse. A bias term was included by augmenting the feature matrix with a column of ones.

The dataset consists of 731 observations, which were randomly split into 80% training data (584 samples) and 20% test data (147 samples) using a fixed random seed for reproducibility. The baseline model was trained using 29 features after preprocessing (including one-hot encoded categorical variables and standardized continuous variables), resulting in 30 learned parameters including the bias.

4.2 Feature Engineering Strategy

To improve predictive performance while retaining interpretability, we introduced non-linear and interaction features in three incremental stages, each motivated by patterns observed during exploratory data analysis. All engineered features were derived exclusively from the training data to avoid data leakage.

Version 1: Initial Non-Linear Features. The baseline model exhibited systematic underfitting, particularly for temperature-related variables. To capture observed curvature and context dependence, we added polynomial terms for temperature and feels-like temperature, along with two interaction terms: temperature $\times$ working day and temperature $\times$ humidity. This increased the feature count from 29 to 33 and yielded substantial reductions in both training and test error.

Added features: temp^2, atemp^2, temp$\times$workingday, temp$\times$humidity.

Version 2: Weather and Seasonal Interactions. To further model context-specific effects, we expanded the feature set by adding quadratic terms for humidity and windspeed and introducing temperature interactions with season, weather situation, and year indicators. These additions allow the model to learn how temperature sensitivity varies across environmental and temporal conditions. The resulting model used 41 features and achieved additional reductions in both training and test RMSE without a large increase in the generalization gap.

Added features: temp^2, atemp^2, hum^2, windspeed^2, $\text{temp}\times\text{season}_{2,3,4}$, $\text{temp}\times\text{weathersit}_{2,3}$, $\text{temp}\times\text{yr}_1$, temp$\times$workingday, temp$\times$humidity.

Version 3: Targeted Higher-Order Interactions. Finally, we introduced a small number of targeted features to capture remaining structure in the data. These included a cubic temperature term to model higher-order curvature, temperature $\times$ holiday interactions, and windspeed $\times$ weather situation interactions to account for varying wind effects under adverse conditions. The final model used 45 features and achieved the lowest test error among all configurations.

Added features: temp^3, temp$\times$holiday, $\text{windspeed}\times\text{weathersit}_{2,3}$.

Version 3 was chosen as the final model because it provided substantial performance gains without violating sample-efficiency constraints. With 45 features and 584 training samples ($\approx$13 samples per feature), the model remained within the commonly accepted 10–15 samples-per-feature range, limiting overfitting risk. Adding further features would have reduced this ratio and increased model variance, offering diminishing returns relative to the added complexity.

5 Results:

Feature_Eng Version	Number of Features	Training RMSE	Test RMSE	Training RMSE Improvement
Baseline	29	742.43	810.93	NIL
Version 1	33	693.34	735.69	12.78 %
Version 2	41	625.72	704.64	15.72 %
Version 3	45	582.87	672.32	21.48%

For the training data, our final model predicts within approximately ±583 bikes on average, while for the test data, predictions are within approximately ±672 bikes. Given an average daily rental count of around 4,500 bikes, this corresponds to roughly 13% training error and 15% test error.

6 Discussion and Conclusion

Our results demonstrate that linear regression can achieve strong predictive performance on bike rental data when combined with thoughtful feature engineering. Non-linear and interaction terms significantly reduced both training and test error, confirming that rental demand is context-dependent and non-linear in nature.

Preprocessing choices such as feature scaling and one-hot encoding improved numerical stability and interpretability. Temperature emerged as the most informative feature, but its predictive power increased substantially when combined with season, weather conditions, and calendar variables.

Despite these improvements, linear regression has inherent limitations. The model can produce negative rental predictions and remains sensitive to multicollinearity, particularly between temperature and feels-like temperature. Additionally, unobserved factors such as special events are not captured in the dataset.

Future work could explore regularized linear models such as Ridge(L2) or Lasso(L1) regression to mitigate multicollinearity and allow the inclusion of a larger feature set without overfitting.We can also add tree-based methods to automatically capture complex non-linear interactions without manual feature engineering. We could evaluate more carefully using cross-validation, time based splits and a separate validation set which can ensure that the results are more reliable and realistic.Nevertheless, our final model provides a strong and interpretable baseline for bike rental demand forecasting.

7 Statement of Contributions

All group members independently completed the full mini-project, including data preprocessing, exploratory data analysis, model implementation, and feature engineering. The final submitted model was selected based on having the lowest test RMSE among the independently developed solutions. Following model selection, all members collaborated jointly on analysis, interpretation of results, figure selection, and writing and editing the final report.

8 Statement on the Use of LLMs and Other Resources

Large language models were used in a limited capacity to assist with clarifying concepts, improving organization, understanding and scrutinizing the visuals in order to help with feature engineering and refining the grammar of the written explanations. for example: We first thought one-hot encoding must be done after the train–test split to avoid data leakage. So, we consulted an LLM and learnt that one-hot encoding doesn't learn anything from the data, it just turns categories into 0s and 1s, so it's safe to do before splitting. All code, analysis, and experimental decisions were developed and brainstormed by the authors.